

```
Processando config 16, seed 0, teste: Alphabit
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config16_seed0.bin
```

```
smultin_MultinomialBitsOver test:
```

```
-----
```

```
N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells =  $2^L = 4$ 
Expected number per cell = 4194304
Hashing = FALSE
```

```
For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
```

```
Delta = 1, Mu = 2.9802323e-08, Sigma = 1
```

```
-----
```

```
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom : 2
Value of the statistic : 1.69
p-value of test : 0.43
```

```
-----
```

```
CPU time used : 00:00:00.01
```

```
Generator state:
```

```
16777216 bits have been read.
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config16_seed0.bin
```

```
smultin_MultinomialBitsOver test:
```

```
-----
```

```
N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells =  $2^L = 16$ 
Expected number per cell = 1048576
Hashing = FALSE
```

```
For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
```

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 8  
Value of the statistic : 7.23  
p-value of test : 0.51

-----  
CPU time used : 00:00:00.01

Generator state:  
33554432 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 128  
Value of the statistic : 120.55  
p-value of test : 0.67

-----  
CPU time used : 00:00:00.00

Generator state:  
50331648 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 33030.71  
p-value of test : 0.15

-----  
CPU time used : 00:00:00.01

Generator state:

67108864 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin

sstring\_HammingIndep test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 177  
Chi-square statistic : 158.11  
p-value of test : 0.84

-----  
CPU time used : 00:00:00.00

Generator state:

83886080 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin

sstring\_HammingIndep test:

-----  
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 305  
Chi-square statistic : 342.31  
p-value of test : 0.07

-----  
CPU time used : 00:00:00.00

Generator state:

100663296 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin

sstring\_HammingCorr test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : -0.15  
p-value of test : 0.56

-----  
CPU time used : 00:00:00.00

Generator state:

117440512 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

Number of degrees of freedom : 32  
ChiSquare statistic : 36.26  
p-value of test : 0.28

-----  
Test on the values of the Statistic M

Number of degrees of freedom : 31  
ChiSquare statistic : 21.01  
p-value of test : 0.91

-----  
Test on the values of the Statistic J

Number of degrees of freedom : 32  
ChiSquare statistic : 25.88  
p-value of test : 0.77

-----  
Test on the values of the Statistic R

Number of degrees of freedom : 25  
ChiSquare statistic : 25.28  
p-value of test : 0.45

-----  
Test on the values of the Statistic C

Number of degrees of freedom : 15  
ChiSquare statistic : 9.83  
p-value of test : 0.83

-----  
CPU time used : 00:00:00.07

Generator state:  
134217728 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config16\_seed0.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

Number of degrees of freedom : 62

ChiSquare statistic : 57.48  
p-value of test : 0.64

-----  
Test on the values of the Statistic M

Number of degrees of freedom : 62  
ChiSquare statistic : 70.77  
p-value of test : 0.21

-----  
Test on the values of the Statistic J

Number of degrees of freedom : 160  
ChiSquare statistic : 165.61  
p-value of test : 0.36

-----  
Test on the values of the Statistic R

Number of degrees of freedom : 57  
ChiSquare statistic : 73.35  
p-value of test : 0.07

-----  
Test on the values of the Statistic C

Number of degrees of freedom : 32  
ChiSquare statistic : 45.49  
p-value of test : 0.06

-----  
CPU time used : 00:00:00.05

Generator state:  
150994688 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3  
Generator: ufile\_CreateReadBin  
Number of bits: 16777216  
Number of statistics: 17  
Total CPU time: 00:00:00.22

All tests were passed